

Поиск файлов. Перенаправление ввода-вывода. Просмотр запущенных процессов

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Цели и задачи работы

Цель лабораторной работы

Ознакомление с инструментами поиска файлов и фильтрации текстовых данных. Приобретение практических навыков: по управлению процессами, по проверке использования диска и обслуживанию файловых систем.

Задачи лабораторной работы

1. Изучить перенаправление ввода-вывода
2. Изучить работу фильтров
3. Изучить команду поиска
4. Ознакомиться с управлением процессами
5. Ознакомиться с командами `df` `du`

Перенаправление ввода-вывода. Работа фильтра

```
foot
[kshved@vbox ~]$ ls /etc > file.txt
[kshved@vbox ~]$ ls ~ >> file.txt
[kshved@vbox ~]$ grep ".conf" file.txt > conf.txt
[kshved@vbox ~]$
```

Рис. 1: выполнение 1-3 задания

```
foot
home/kshved/work/study/2024-2025/Операционные системы/os-Intro/project-personal/stage1/report/bib/cite.bib
home/kshved/work/study/2024-2025/Операционные системы/os-Intro/project-personal/stage1/report/pandoc/filters/pandocxnos/core.py
home/kshved/work/study/2024-2025/Операционные системы/os-Intro/project-personal/stage2/report/bib/cite.bib
home/kshved/work/study/2024-2025/Операционные системы/os-Intro/project-personal/stage2/report/pandoc/filters/pandocxnos/core.py
home/kshved/work/study/2024-2025/Операционные системы/os-Intro/project-personal/stage3/report/bib/cite.bib
home/kshved/work/study/2024-2025/Операционные системы/os-Intro/project-personal/stage3/report/pandoc/filters/pandocxnos/core.py
home/kshved/work/study/2024-2025/Операционные системы/os-Intro/project-personal/stage4/report/bib/cite.bib
home/kshved/work/study/2024-2025/Операционные системы/os-Intro/project-personal/stage4/report/pandoc/filters/pandocxnos/core.py
home/kshved/work/study/2024-2025/Операционные системы/os-Intro/project-personal/stage5/report/bib/cite.bib
home/kshved/work/study/2024-2025/Операционные системы/os-Intro/project-personal/stage5/report/pandoc/filters/pandocxnos/core.py
home/kshved/work/study/2024-2025/Операционные системы/os-Intro/project-personal/stage6/report/bib/cite.bib
home/kshved/work/study/2024-2025/Операционные системы/os-Intro/project-personal/stage6/report/pandoc/filters/pandocxnos/core.py
home/kshved/work/study/2024-2025/Операционные системы/os-Intro/presentation/report/bib/cite.bib
home/kshved/work/study/2024-2025/Операционные системы/os-Intro/presentation/report/pandoc/filters/pandocxnos/core.py
home/kshved/work/study/2024-2025/Операционные системы/git-extended/.git/hooks/commit-msg.sample
home/kshved/work/study/2024-2025/Операционные системы/git-extended/.git/objects/39/c6b93b98Feb32f59888a534922cf8c83316a37
home/kshved/work/study/2024-2025/Операционные системы/git-extended/.git/config
home/kshved/.password-store/.git/hooks/commit-msg.sample
home/kshved/.password-store/.git/config
home/kshved/chezmoi.2.68.1_linux_and64.tar.gz
home/kshved/completions/chezmoi-completion.bash
home/kshved/completions/chezmoi.fish
home/kshved/completions/chezmoi.ps1
home/kshved/completions/chezmoi.zsh
home/kshved/conf.txt
kshved@vbox ~$ ls -l | grep "c"
chezmoi.2.68.1_linux_and64.tar.gz
completions
conf.txt
kshved@vbox ~$
```

Рис. 2: выполнение 4 задания

```
host.conf  
hostname  
hosts  
(END)
```

A terminal window with a dark background and light-colored text. The prompt is [kshved@vbox ~]. The commands and their outputs are: 1. ls /etc | grep "^h" | less (output: Mar 27), 2. find ~ -name "log*" > ~/logfile & (output: [1] 5044), 3. rm ~/logfile (output: [1]+ Завершён find ~ -name "log*" > ~/logfile Mar 25), 4. The prompt returns to [kshved@vbox ~]\$ with a cursor. There is a faint "поняла" watermark in the bottom right of the terminal area.

```
[kshved@vbox ~]$ ls /etc | grep "^h" | less
[kshved@vbox ~]$ find ~ -name "log*" > ~/logfile &
[1] 5044
[kshved@vbox ~]$ rm ~/logfile
[1]+ Завершён find ~ -name "log*" > ~/logfile
[kshved@vbox ~]$
```

Рис. 4: выполнение 6-7 задания

Команды df и du

```
~ /home/kshvedy.com - ssh
1213852 /home/kshved
[kshved@vbox ~]$ df
Файловая система 1К-блоков  Использовано  Доступно  Использованой  Смонтировано в
/dev/sda3          32199680      9141592  21583648         38% /
devtmpfs           4896             0      4896             0% /dev
tmpfs              4365564         12    4365552         1% /dev/shm
tmpfs              1746228         1332    1744896         1% /run
tmpfs              1824             0      1824             0% /run/credentials/systemd-network-generator.service
tmpfs              1824             0      1824             0% /run/credentials/systemd-journald.service
tmpfs              1824             0      1824             0% /run/credentials/systemd-udev-load-credentials.service
tmpfs              1824             0      1824             0% /run/credentials/systemd-sysctl.service
tmpfs              1824             0      1824             0% /run/credentials/systemd-tapfiles-setup-dev-early.service
tmpfs              1824             0      1824             0% /run/credentials/systemd-tapfiles-setup-dev.service
tmpfs              4365564         16    4365548         1% /tmp
/dev/sda3          32199680      9141592  21583648         38% /home
tmpfs              1824             0      1824             0% /run/credentials/systemd-vconsole-setup.service
/dev/sda2          996780         388360    547688         41% /boot
tmpfs              1824             0      1824             0% /run/credentials/systemd-tapfiles-setup.service
tmpfs              1824             0      1824             0% /run/credentials/systemd-resolved.service
tmpfs              873112         188    872924         1% /run/user/1000
```

Рис. 5: выполнение команды df

Команды df и du

```
foot
/home/kshved/.var/app/org.telegram.desktop/TelegramDesktop/tdata/user_data/cache/8/56/2A7E1EACEC49
/home/kshved/.var/app/org.telegram.desktop/data/TelegramDesktop/tdata/user_data/cache/8/56/65E4A18B1471
24 /home/kshved/.var/app/org.telegram.desktop/data/TelegramDesktop/tdata/user_data/cache/8/56/15541C76144A
84 /home/kshved/.var/app/org.telegram.desktop/data/TelegramDesktop/tdata/user_data/cache/8/56/D0B182E7C91
24 /home/kshved/.var/app/org.telegram.desktop/data/TelegramDesktop/tdata/user_data/cache/8/56/05E839C45C78
132 /home/kshved/.var/app/org.telegram.desktop/data/TelegramDesktop/tdata/user_data/cache/8/56/3C4C4742E512
32 /home/kshved/.var/app/org.telegram.desktop/data/TelegramDesktop/tdata/user_data/cache/8/56/A557B4393DC
388 /home/kshved/.var/app/org.telegram.desktop/data/TelegramDesktop/tdata/user_data/cache/8/56
8 /home/kshved/.var/app/org.telegram.desktop/data/TelegramDesktop/tdata/user_data/cache/8/AS/3F58CACBD8
4 /home/kshved/.var/app/org.telegram.desktop/data/TelegramDesktop/tdata/user_data/cache/8/AS/6AC5CFF8BE
48 /home/kshved/.var/app/org.telegram.desktop/data/TelegramDesktop/tdata/user_data/cache/8/AS/58A83C8A6181
16 /home/kshved/.var/app/org.telegram.desktop/data/TelegramDesktop/tdata/user_data/cache/8/AS/569A3A1982DF
88 /home/kshved/.var/app/org.telegram.desktop/data/TelegramDesktop/tdata/user_data/cache/8/AS
12 /home/kshved/.var/app/org.telegram.desktop/data/TelegramDesktop/tdata/user_data/cache/8/ES/C5CA21465894
52 /home/kshved/.var/app/org.telegram.desktop/data/TelegramDesktop/tdata/user_data/cache/8/ES/33F95800C878
132 /home/kshved/.var/app/org.telegram.desktop/data/TelegramDesktop/tdata/user_data/cache/8/ES/E241857784AE
8 /home/kshved/.var/app/org.telegram.desktop/data/TelegramDesktop/tdata/user_data/cache/8/ES/819300C3A8B
76 /home/kshved/.var/app/org.telegram.desktop/data/TelegramDesktop/tdata/user_data/cache/8/ES/CC38A632C3C7
76 /home/kshved/.var/app/org.telegram.desktop/data/TelegramDesktop/tdata/user_data/cache/8/ES/2E444288A2B8
356 /home/kshved/.var/app/org.telegram.desktop/data/TelegramDesktop/tdata/user_data/cache/8/ES
28 /home/kshved/.var/app/org.telegram.desktop/data/TelegramDesktop/tdata/user_data/cache/8/8E/58A4180E68D3
64 /home/kshved/.var/app/org.telegram.desktop/data/TelegramDesktop/tdata/user_data/cache/8/8E/482A88CCADF
12 /home/kshved/.var/app/org.telegram.desktop/data/TelegramDesktop/tdata/user_data/cache/8/8E/DF8307D1C958
12 /home/kshved/.var/app/org.telegram.desktop/data/TelegramDesktop/tdata/user_data/cache/8/8E/483789F8F227
188 /home/kshved/.var/app/org.telegram.desktop/data/TelegramDesktop/tdata/user_data/cache/8/8E
12 /home/kshved/.var/app/org.telegram.desktop/data/TelegramDesktop/tdata/user_data/cache/8/F8/874882344E9F
12 /home/kshved/.var/app/org.telegram.desktop/data/TelegramDesktop/tdata/user_data/cache/8/F8/587FDC78E7FB
16 /home/kshved/.var/app/org.telegram.desktop/data/TelegramDesktop/tdata/user_data/cache/8/F8/668F79E273CF
48 /home/kshved/.var/app/org.telegram.desktop/data/TelegramDesktop/tdata/user_data/cache/8/F8
12 /home/kshved/.var/app/org.telegram.desktop/data/TelegramDesktop/tdata/user_data/cache/8/E1/05747CA5A86
```

Рис. 6: выполнение команды du -a ~


```
fat
/home/sshwd/.local/share/pnpm/global/.pnpm/fast-uri@13.8.6/node_modules
/home/sshwd/.local/share/pnpm/global/.pnpm/fast-uri@13.8.6/node_modules/fast-uri
/home/sshwd/.local/share/pnpm/global/.pnpm/fast-uri@13.8.6/node_modules/fast-uri/lib
/home/sshwd/.local/share/pnpm/global/.pnpm/fast-uri@13.8.6/node_modules/fast-uri/test
/home/sshwd/.local/share/pnpm/global/.pnpm/fast-uri@13.8.6/node_modules/fast-uri/types
/home/sshwd/.local/share/pnpm/global/.pnpm/fast-uri@13.8.6/node_modules/fast-uri/.github
/home/sshwd/.local/share/pnpm/global/.pnpm/fast-uri@13.8.6/node_modules/fast-uri/.github/workflows
/home/sshwd/.local/share/pnpm/global/.pnpm/comiconfig@9.8.8.typescript@5.8.2
/home/sshwd/.local/share/pnpm/global/.pnpm/comiconfig@9.8.8.typescript@5.8.2/node_modules
/home/sshwd/.local/share/pnpm/global/.pnpm/comiconfig@9.8.8.typescript@5.8.2/node_modules/comiconfig
/home/sshwd/.local/share/pnpm/global/.pnpm/comiconfig@9.8.8.typescript@5.8.2/node_modules/comiconfig/dist
/home/sshwd/.local/share/pnpm/global/.pnpm/comiconfig@9.8.8.typescript@5.8.2/node_modules/comiconfig/node_modules
/home/sshwd/.local/share/pnpm/global/.pnpm/comiconfig@9.8.8.typescript@5.8.2/node_modules/comiconfig/node_modules/.bin
/home/sshwd/.local/share/pnpm/global/.pnpm/eno-path@2.2.1
/home/sshwd/.local/share/pnpm/global/.pnpm/eno-path@2.2.1/node_modules
/home/sshwd/.local/share/pnpm/global/.pnpm/eno-path@2.2.1/node_modules/eno-path
/home/sshwd/.local/share/pnpm/global/.pnpm/js-yaml@4.1.8
/home/sshwd/.local/share/pnpm/global/.pnpm/js-yaml@4.1.8/node_modules
/home/sshwd/.local/share/pnpm/global/.pnpm/js-yaml@4.1.8/node_modules/js-yaml
/home/sshwd/.local/share/pnpm/global/.pnpm/js-yaml@4.1.8/node_modules/js-yaml/lib
/home/sshwd/.local/share/pnpm/global/.pnpm/js-yaml@4.1.8/node_modules/js-yaml/lib/type
/home/sshwd/.local/share/pnpm/global/.pnpm/js-yaml@4.1.8/node_modules/js-yaml/lib/schema
/home/sshwd/.local/share/pnpm/global/.pnpm/js-yaml@4.1.8/node_modules/js-yaml/dist
/home/sshwd/.local/share/pnpm/global/.pnpm/js-yaml@4.1.8/node_modules/js-yaml/dist
/home/sshwd/.local/share/pnpm/global/.pnpm/js-yaml@4.1.8/node_modules/js-yaml/node_modules
/home/sshwd/.local/share/pnpm/global/.pnpm/js-yaml@4.1.8/node_modules/js-yaml/node_modules/.bin
/home/sshwd/.local/share/pnpm/global/.pnpm/argparse@2.8.1
/home/sshwd/.local/share/pnpm/global/.pnpm/argparse@2.8.1/node_modules
/home/sshwd/.local/share/pnpm/global/.pnpm/argparse@2.8.1/node_modules/argparse
```

Рис. 7: выполнение команды find

В данной работе мы ознакомились с инструментами поиска файлов и фильтрации текстовых данных. А также приобрели практические навыки по управлению процессами.